Deep Generative Models

Lecture 4

Roman Isachenko

AI Masters

2026, Spring

Recap of Previous Lecture

MLE for LVM

$$\boldsymbol{\theta}^* = \arg\max_{\boldsymbol{\theta}} \sum_{i=1}^{n} \log p_{\boldsymbol{\theta}}(\mathbf{x}_i) = \arg\max_{\boldsymbol{\theta}} \sum_{i=1}^{n} \log \int p_{\boldsymbol{\theta}}(\mathbf{x}_i|\mathbf{z}_i)p(\mathbf{z}_i)d\mathbf{z}_i.$$

Naive Monte Carlo Estimation

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) = \log \mathbb{E}_{p(\mathbf{z})} p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) \geq \mathbb{E}_{p(\mathbf{z})} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) \approx \frac{1}{K}\sum_{k=1}^{K} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}_k),$$

where $\mathbf{z}_k \sim p(\mathbf{z})$.

Variational Lower Bound (ELBO)

$$\mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_q \log \frac{p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})}$$

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) = \mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}) + D_{\mathrm{KL}}(q(\mathbf{z}) \| p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})) \geq \mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}).$$

Recap of Previous Lecture

Variational Evidence Lower Bound (ELBO)

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) = \mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}) + D_{\mathrm{KL}}(q(\mathbf{z}) \| p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})) \geq \mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}).$$

$$\mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}) = \int q(\mathbf{z}) \log \frac{p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z})} d\mathbf{z} = \mathbb{E}_q \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - D_{\mathrm{KL}}(q(\mathbf{z}) \| p(\mathbf{z}))$$

Log-likelihood Decomposition

$$\log p_{\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_q \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - D_{\mathrm{KL}}(q(\mathbf{z}) \| p(\mathbf{z})) + D_{\mathrm{KL}}(q(\mathbf{z}) \| p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})).$$

- Rather than maximizing likelihood, maximize the ELBO:

$$\max_{\boldsymbol{\theta}} p_{\boldsymbol{\theta}}(\mathbf{x}) \quad \rightarrow \quad \max_{q,\boldsymbol{\theta}} \mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x})$$

- Maximizing the ELBO with respect to the variational distribution q is equivalent to minimizing the KL divergence:

$$\arg\max_q \mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}) \equiv \arg\min_q D_{\mathrm{KL}}(q(\mathbf{z}) \| p_{\boldsymbol{\theta}}(\mathbf{z}|\mathbf{x})).$$

Recap of Previous Lecture

$$
\begin{aligned}
\mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}) &= \mathbb{E}_q \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - D_{\mathrm{KL}}(q(\mathbf{z}) \| p(\mathbf{z})) = \\
&= \mathbb{E}_q \left[\log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - \log \frac{q(\mathbf{z})}{p(\mathbf{z})} \right] d\mathbf{z} \rightarrow \max_{q,\boldsymbol{\theta}}.
\end{aligned}
$$

Variational Posterior

$$
\begin{aligned}
q^*(\mathbf{z}) &= \arg\max_q \mathcal{L}_{q,\boldsymbol{\theta}^*}(\mathbf{x}) = \\
&= \arg\min_q D_{\mathrm{KL}}(q(\mathbf{z}) \| p_{\boldsymbol{\theta}^*}(\mathbf{z}|\mathbf{x})) = p_{\boldsymbol{\theta}^*}(\mathbf{z}|\mathbf{x});
\end{aligned}
$$

Amortized Variational Inference

We restrict the family of possible distributions $q(\mathbf{z})$ to a parametric class $q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})$, conditioned on data $\mathbf{x}$ and parameterized by $\boldsymbol{\phi}$.

Gradient Update

$$
\begin{bmatrix} \boldsymbol{\phi}_k \\ \boldsymbol{\theta}_k \end{bmatrix} = \left.\begin{bmatrix} \boldsymbol{\phi}_{k-1} + \eta \cdot \nabla_{\boldsymbol{\phi}} \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) \\ \boldsymbol{\theta}_{k-1} + \eta \cdot \nabla_{\boldsymbol{\theta}} \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) \end{bmatrix}\right|_{(\boldsymbol{\phi}_{k-1}, \boldsymbol{\theta}_{k-1})}
$$

Recap of Previous Lecture

$$\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})) \to \max_{\boldsymbol{\phi},\boldsymbol{\theta}}.$$

Gradient w.r.t. $\boldsymbol{\theta}$ — Monte Carlo Estimation

$$\nabla_{\boldsymbol{\theta}} \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) = \int q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}) \nabla_{\boldsymbol{\theta}} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) d\mathbf{z} \approx$$

$$\approx \nabla_{\boldsymbol{\theta}} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}^*), \quad \mathbf{z}^* \sim q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}).$$

Gradient w.r.t. $\boldsymbol{\phi}$ — Reparameterization Trick

$$\nabla_{\boldsymbol{\phi}} \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) = \int p(\boldsymbol{\epsilon}) \nabla_{\boldsymbol{\phi}} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{g}_{\boldsymbol{\phi}}(\mathbf{x}, \boldsymbol{\epsilon})) d\boldsymbol{\epsilon} - \nabla_{\boldsymbol{\phi}} D_{\mathrm{KL}}$$

$$\approx \nabla_{\boldsymbol{\phi}} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{g}_{\boldsymbol{\phi}}(\mathbf{x}, \boldsymbol{\epsilon}^*)) - \nabla_{\boldsymbol{\phi}} \mathrm{KL}$$

Variational Assumption

$$p(\boldsymbol{\epsilon}) = \mathcal{N}(0, \mathbf{I}); \quad q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{\phi}}(\mathbf{x}), \boldsymbol{\sigma}^2_{\boldsymbol{\phi}}(\mathbf{x})).$$

$$\mathbf{z} = \mathbf{g}_{\boldsymbol{\phi}}(\mathbf{x}, \boldsymbol{\epsilon}) = \boldsymbol{\sigma}_{\boldsymbol{\phi}}(\mathbf{x}) \odot \boldsymbol{\epsilon} + \boldsymbol{\mu}_{\boldsymbol{\phi}}(\mathbf{x}).$$

Recap of Previous Lecture

Training

- Pick a batch of samples $\{\mathbf{x}_i\}_{i=1}^{B}$ (here we use Monte Carlo technique).
- Compute the objective for each sample (apply the reparametrization trick):

$$\boldsymbol{\epsilon}^* \sim p(\boldsymbol{\epsilon}); \quad \mathbf{z}^* = \mathbf{g}_{\boldsymbol{\phi}}(\mathbf{x}, \boldsymbol{\epsilon}^*);$$

$$\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) \approx \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}^*) - D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})).$$

- Update parameters via stochastic gradient steps with respect to $\boldsymbol{\phi}$ and $\boldsymbol{\theta}$.

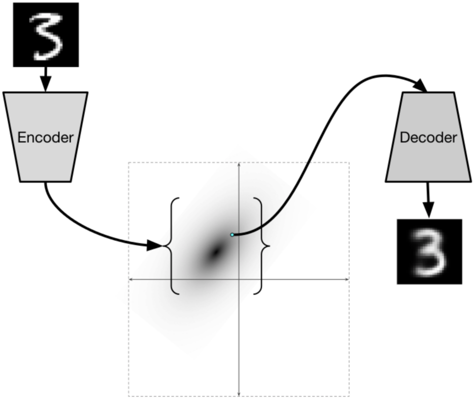
Inference

- Sample $\mathbf{z}^*$ from the prior $p(\mathbf{z})$ ($\mathcal{N}(0, \mathbf{I})$);
- Generate data from the decoder $p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}^*)$.

Note: The encoder $q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})$ isn't needed during generation.

Recap of Previous Lecture

$$\mathcal{L}_{q,\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_q \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - D_{\mathrm{KL}}(q_{\phi}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))$$



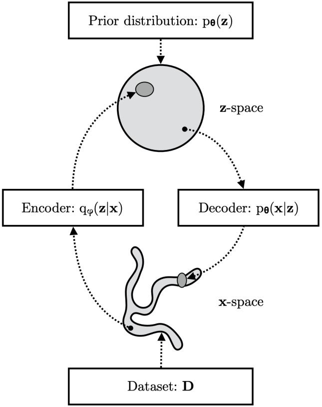


image credit: http://ijdykeman.github.io/ml/2016/12/21/cvae.html
Kingma D. P., Welling M. An Introduction to Variational Autoencoders, 2019

Outline

1. ELBO Surgery
2. Discrete VAE Latent Representations
3. Vector Quantized VAE
4. Likelihood-Free Learning

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ELBO Surgery

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Hoffman M. D., Johnson M. J. ELBO Surgery: Yet Another Way to Carve Up the Variational Evidence Lower Bound, 2016

ELBO Surgery

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Theorem

$$\frac{1}{n}\sum_{i=1}^{n}D_{\mathrm{KL}}(q_\phi(\mathbf{z}|\mathbf{x}_i)\|p(\mathbf{z})) = D_{\mathrm{KL}}(q_{\mathrm{agg},\phi}(\mathbf{z})\|p(\mathbf{z})) + \mathbb{I}_q[\mathbf{x},\mathbf{z}];$$

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ELBO Surgery

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Theorem

$$\frac{1}{n}\sum_{i=1}^{n}D_{\mathrm{KL}}(q_\phi(\mathbf{z}|\mathbf{x}_i)\|p(\mathbf{z})) = D_{\mathrm{KL}}(q_{\mathrm{agg},\phi}(\mathbf{z})\|p(\mathbf{z})) + \mathbb{I}_q[\mathbf{x},\mathbf{z}];$$

- $q_{\mathrm{agg},\phi}(\mathbf{z}) = \frac{1}{n}\sum_{i=1}^{n} q_\phi(\mathbf{z}|\mathbf{x}_i)$ denotes the **aggregated** variational posterior.
- $\mathbb{I}_q[\mathbf{x},\mathbf{z}]$ is the mutual information between $\mathbf{x}$ and $\mathbf{z}$ under the data distribution $p_{\mathrm{data}}(\mathbf{x})$ and $q_\phi(\mathbf{z}|\mathbf{x})$.
- The first term encourages $q_{\mathrm{agg},\phi}(\mathbf{z})$ to match the prior $p(\mathbf{z})$.
- The second term reduces the information about $\mathbf{x}$ encoded in $\mathbf{z}$.

Hoffman M. D., Johnson M. J. ELBO Surgery: Yet Another Way to Carve Up the Variational Evidence Lower Bound, 2016

ELBO Surgery

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Proof

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$$= D_{\mathrm{KL}}(q_{\mathrm{agg},\phi}(\mathbf{z})\|p(\mathbf{z})) + \frac{1}{n}\sum_{i=1}^{n} D_{\mathrm{KL}}(q_\phi(\mathbf{z}|\mathbf{x}_i)\|q_{\mathrm{agg},\phi}(\mathbf{z}))$$

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$$= D_{\mathrm{KL}}(q_{\mathrm{agg},\phi}(\mathbf{z})\|p(\mathbf{z})) + \frac{1}{n}\sum_{i=1}^{n} D_{\mathrm{KL}}(q_\phi(\mathbf{z}|\mathbf{x}_i)\|q_{\mathrm{agg},\phi}(\mathbf{z}))$$

$$\mathbb{I}_q[\mathbf{x},\mathbf{z}] = \frac{1}{n}\sum_{i=1}^{n} D_{\mathrm{KL}}(q_\phi(\mathbf{z}|\mathbf{x}_i)\|q_{\mathrm{agg},\phi}(\mathbf{z})).$$

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ELBO Surgery

Revisiting the ELBO

$$\frac{1}{n}\sum_{i=1}^{n}\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}_i) = \frac{1}{n}\sum_{i=1}^{n}\left[\mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}_i)}\log p_{\boldsymbol{\theta}}(\mathbf{x}_i|\mathbf{z}) - D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}_i)\|p(\mathbf{z}))\right]$$

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$$= \underbrace{\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x}_i)}\log p_{\boldsymbol{\theta}}(\mathbf{x}_i|\mathbf{z})}_{\text{Reconstruction Loss}} - \underbrace{\mathbb{I}_q[\mathbf{x},\mathbf{z}]}_{\text{Mutual Information}} - \underbrace{D_{\mathrm{KL}}(q_{\mathrm{agg},\phi}(\mathbf{z})\|p(\mathbf{z}))}_{\text{Marginal KL}}$$

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The prior distribution $p(\mathbf{z})$ only appears in the last term.

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ELBO Surgery

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$$\frac{1}{n}\sum_{i=1}^{n}\mathcal{L}_{\phi,\boldsymbol{\theta}}(\mathbf{x}_i) = \frac{1}{n}\sum_{i=1}^{n}\left[\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x}_i)}\log p_{\boldsymbol{\theta}}(\mathbf{x}_i|\mathbf{z}) - D_{\mathrm{KL}}(q_\phi(\mathbf{z}|\mathbf{x}_i)\|p(\mathbf{z}))\right] =$$

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The prior distribution $p(\mathbf{z})$ only appears in the last term.

Optimal VAE Prior

$$D_{\mathrm{KL}}(q_{\mathrm{agg},\phi}(\mathbf{z})\|p(\mathbf{z})) = 0 \quad \Leftrightarrow \quad p(\mathbf{z}) = q_{\mathrm{agg},\phi}(\mathbf{z}) = \frac{1}{n}\sum_{i=1}^{n} q_\phi(\mathbf{z}|\mathbf{x}_i).$$

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Hence, the optimal prior $p(\mathbf{z})$ is the aggregated variational posterior $q_{\text{agg},\phi}(\mathbf{z})$.

Hoffman M. D., Johnson M. J. ELBO Surgery: Yet Another Way to Carve Up the Variational Evidence Lower Bound, 2016

Marginal KL

$$D_{\mathrm{KL}}(q_{\text{agg},\phi}(\mathbf{z}) \| p(\mathbf{z}))$$

- $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_\phi(\mathbf{x}), \boldsymbol{\sigma}^2_\phi(\mathbf{x}))$ is unimodal.
- It is generally believed that the **mismatch between $p(\mathbf{z})$ and $q_{\text{agg},\phi}(\mathbf{z})$** is the primary explanation for blurry VAE-generated images.

Rezende D. J., Mohamed S. Variational Inference with Normalizing Flows, 2015

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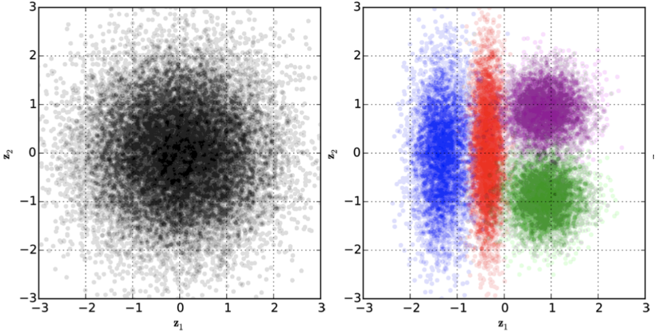

(a) Prior distribution

(b) Posteriors in standard VAE

Rezende D. J., Mohamed S. Variational Inference with Normalizing Flows, 2015

Why are VAE Generations Blurry?

Consider a Gaussian decoder $p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{\theta}}(\mathbf{z}), \sigma^2\mathbf{I})$.

Lai C. H. et al. The principles of diffusion models, 2025.

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ELBO Optimization

With **fixed** encoder $q_{\phi}(\mathbf{z}|\mathbf{x})$, optimizing ELBO reduces to

$$\arg\min_{\boldsymbol{\theta}} \mathbb{E}_{p_{\text{data}}(\mathbf{x})q_{\phi}(\mathbf{z}|\mathbf{x})} \|\mathbf{x} - \boldsymbol{\mu}_{\boldsymbol{\theta}}(\mathbf{z})\|^2$$

The optimal decoder mean is the **conditional expectation**:

$$\boldsymbol{\mu}^*(\mathbf{z}) = \mathbb{E}_{q_{\phi}(\mathbf{x}|\mathbf{z})}[\mathbf{x}] = \frac{\mathbb{E}_{p_{\text{data}}(\mathbf{x})}[q_{\phi}(\mathbf{z}|\mathbf{x}) \cdot \mathbf{x}]}{\mathbb{E}_{p_{\text{data}}(\mathbf{x})}[q_{\phi}(\mathbf{z}|\mathbf{x})]}$$

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The optimal decoder mean is the **conditional expectation**:

$$\boldsymbol{\mu}^*(\mathbf{z}) = \mathbb{E}_{q_{\phi}(\mathbf{x}|\mathbf{z})}[\mathbf{x}] = \frac{\mathbb{E}_{p_{\text{data}}(\mathbf{x})}[q_{\phi}(\mathbf{z}|\mathbf{x}) \cdot \mathbf{x}]}{\mathbb{E}_{p_{\text{data}}(\mathbf{x})}[q_{\phi}(\mathbf{z}|\mathbf{x})]}$$

Blurriness from Averaging

- $p_{\text{data}}(\mathbf{x})$ and $q_{\phi}(\mathbf{z}|\mathbf{x})$ give us the aggregated posterior $q_{\text{agg},\phi}(\mathbf{z})$.
- If $\mathbf{x} \neq \mathbf{x}'$ map to overlapping latent regions, $\boldsymbol{\mu}^*(\mathbf{z})$ averages over unrelated inputs.
- This leads to **blurry, non-distinct outputs**.

Lai C. H. et al. The principles of diffusion models, 2025.

Outline

1. ELBO Surgery
2. Discrete VAE Latent Representations
3. Vector Quantized VAE
4. Likelihood-Free Learning

Discrete VAE Latents

Motivation

- VAE has used **continuous** latent variables $\mathbf{z}$.
- For some modalities, **discrete** representations $\mathbf{z}$ may be a more natural choice.
- Advanced autoregressive models are highly effective for distributions over discrete variables (for example, transformers process discrete tokens).

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ELBO

$$\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) - D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})) \rightarrow \max_{\boldsymbol{\phi},\boldsymbol{\theta}}.$$

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- Apply the reparametrization trick to obtain unbiased gradients.
- Use Gaussian distributions for $q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x})$ and $p(\mathbf{z})$ to compute the KL analytically.

Discrete VAE Latents

Assumptions

- Let $c \sim \mathrm{Cat}(\boldsymbol{\pi})$, where
$$\boldsymbol{\pi} = (\pi_1, \ldots, \pi_K), \quad \pi_k = P(c = k), \quad \sum_{k=1}^{K} \pi_k = 1.$$
- Suppose the VAE adopts a discrete latent variable c with prior $p(c) = \mathrm{Uniform}\{1, \ldots, K\}$.

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$$= -\mathrm{H}(q_{\boldsymbol{\phi}}(c|\mathbf{x})) + \log K.$$

Discrete VAE Latents

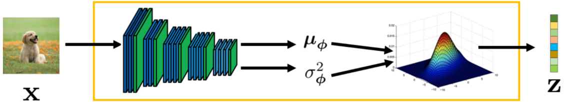
$$\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_{q_{\boldsymbol{\phi}}(c|\mathbf{x})} \log p_{\boldsymbol{\theta}}(\mathbf{x}|c) + \mathrm{H}(q_{\boldsymbol{\phi}}(c|\mathbf{x})) - \log K \rightarrow \max_{\boldsymbol{\phi},\boldsymbol{\theta}}.$$

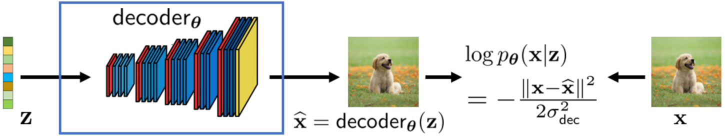
Chan S., Tutorial on Diffusion Models for Imaging and Vision, 2024

Discrete VAE Latents

$$\mathcal{L}_{\phi,\theta}(\mathbf{x}) = \mathbb{E}_{q_\phi(c|\mathbf{x})} \log p_\theta(\mathbf{x}|c) + \mathrm{H}(q_\phi(c|\mathbf{x})) - \log K \to \max_{\phi,\theta}.$$

- The encoder should output a discrete distribution $q_\phi(c|\mathbf{x})$.
- We need an analogue of the reparametrization trick for discrete $q_\phi(c|\mathbf{x})$.
- The decoder $p_\theta(\mathbf{x}|c)$ must take a discrete random variable c as input.





Chan S., Tutorial on Diffusion Models for Imaging and Vision, 2024

Outline

Vector Quantization

Define the codebook (dictionary) space $\{\mathbf{e}_k\}_{k=1}^{K}$ with $\mathbf{e}_k \in \mathbb{R}^L$ and K the number of codebook entries.

Zhao Y. et al. Feature Quantization Improves GAN Training, 2020

Vector Quantization

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Quantized Representation

A quantized vector $\mathbf{z}_q \in \mathbb{R}^L$, for any $\mathbf{z} \in \mathbb{R}^L$, is defined via nearest-neighbor lookup in the codebook:

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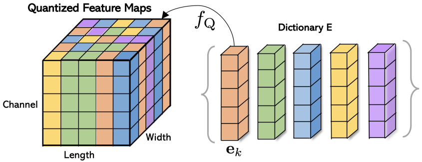
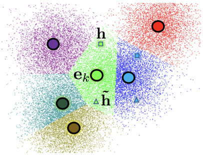
Quantization Procedure

If the encoded tensor has spatial dimensions, quantization is independently applied to each of the $W \times H$ locations.





Zhao Y. et al. Feature Quantization Improves GAN Training, 2020

Vector Quantized VAE (VQ-VAE)

- The encoder outputs a continuous vector $\mathbf{z}_e = \mathrm{NN}_{e,\phi}(\mathbf{x}) \in \mathbb{R}^L$.
- Quantization deterministically maps $\mathbf{z}_e$ to its quantized codebook vector $\mathbf{z}_q$.
- The decoder is conditioned on codebook entries $\mathbf{e}_c$, i.e., via $p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{e}_c)$ (instead of $p_{\boldsymbol{\theta}}(\mathbf{x}|c)$).

Oord A., Vinyals O., Kavukcuoglu K. Neural Discrete Representation Learning, 2017

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Deterministic Variational Posterior

$$q_\phi(c = k^*|\mathbf{x}) = \begin{cases} 1, & \text{for } k^* = \arg\min_k \|\mathbf{z}_e - \mathbf{e}_k\|; \\ 0, & \text{otherwise.} \end{cases}$$

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Note: The KL regularizer becomes constant and has no direct effect on the ELBO objective in this case.

Oord A., Vinyals O., Kavukcuoglu K. Neural Discrete Representation Learning, 2017

Vector Quantized VAE (VQ-VAE): Forward

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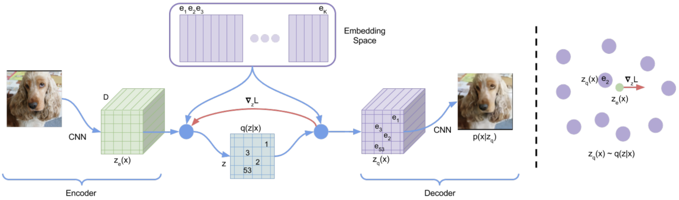

Oord A., Vinyals O., Kavukcuoglu K. Neural Discrete Representation Learning, 2017

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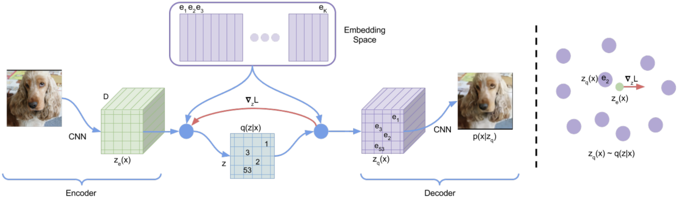

Challenge: The $\arg\min$ operation is non-differentiable.

Oord A., Vinyals O., Kavukcuoglu K. Neural Discrete Representation Learning, 2017

Vector Quantized VAE (VQ-VAE): Backward

ELBO

$$\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) = \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}_q) - \log K, \quad \mathbf{z}_q = \mathbf{e}_{k^*}, \; k^* = \arg\min_k \|\mathbf{z}_e - \mathbf{e}_k\|.$$

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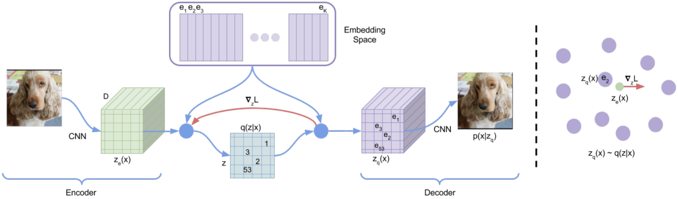

Oord A., Vinyals O., Kavukcuoglu K. Neural Discrete Representation Learning, 2017

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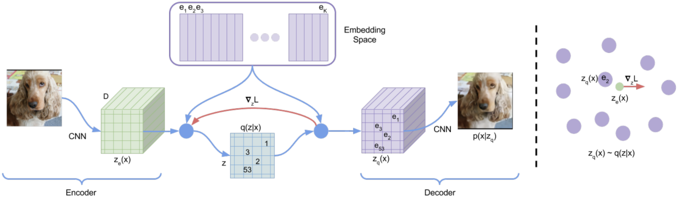

Straight-Through Gradient Estimator

$$\frac{\partial \log p(\mathbf{x}|\mathbf{z}_q, \boldsymbol{\theta})}{\partial \phi} = \frac{\partial \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}_q)}{\partial \mathbf{z}_q} \cdot \color{red}{\frac{\partial \mathbf{z}_q}{\partial \phi}} =$$

Oord A., Vinyals O., Kavukcuoglu K. Neural Discrete Representation Learning, 2017

Vector Quantized VAE (VQ-VAE): Backward

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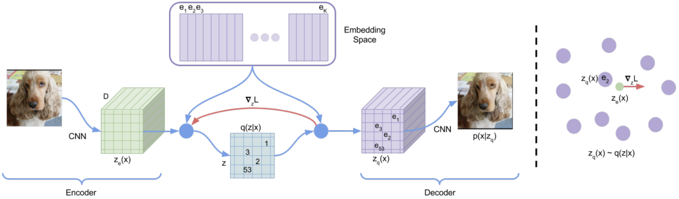

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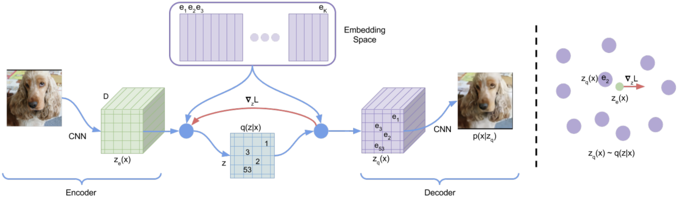
$$= \frac{\partial \log p_{\theta}(\mathbf{x}|\mathbf{z}_q)}{\partial \mathbf{z}_q} \cdot \frac{\partial \mathbf{z}_q}{\partial \mathbf{z}_e} \cdot \frac{\partial \mathbf{z}_e}{\partial \phi}$$

Oord A., Vinyals O., Kavukcuoglu K. Neural Discrete Representation Learning, 2017

Vector Quantized VAE (VQ-VAE): Backward

ELBO

$$\mathcal{L}_{\phi,\boldsymbol{\theta}}(\mathbf{x}) = \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}_q) - \log K, \quad \mathbf{z}_q = \mathbf{e}_{k^*}, \ k^* = \arg\min_k \|\mathbf{z}_e - \mathbf{e}_k\|.$$



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Oord A., Vinyals O., Kavukcuoglu K. Neural Discrete Representation Learning, 2017

Vector Quantized VAE-2 (VQ-VAE-2)

Extension to the spatial domain: $\mathbf{c} \in \{1, \dots, K\}^{W \times H}$

$$q_\phi(\mathbf{c}|\mathbf{x}) = \prod_{i=1}^{W} \prod_{j=1}^{H} q(c_{ij}|\mathbf{x}, \phi); \quad p(\mathbf{c}) = \prod_{i=1}^{W} \prod_{j=1}^{H} \text{Uniform}\{1, \dots, K\}.$$

Sample Diversity

VQ-VAE (Proposed) **BigGAN deep**

Razavi A., Oord A., Vinyals O. Generating Diverse High-Fidelity Images with VQ-VAE-2, 2019

Vector Quantized VAE (VQ-VAE): Final algorithm

Training

- Vector-quantize (per spatial location if applicable):

$$k^* = \arg\min_k \|\mathbf{z}_e - \mathbf{e}_k\|_2, \quad \mathbf{z}_q = \mathbf{e}_{k^*}, \quad \mathbf{z}_e = \mathrm{NN}_{e,\phi}(\mathbf{x}).$$

- Compute ELBO objective:

$$\mathcal{L}_{\phi,\boldsymbol{\theta}}(\mathbf{x}) = \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}_q) - \log K.$$

- Compute total loss with codebook and commitment losses (with stop-gradient $\mathrm{sg}[\cdot]$):

$$\mathcal{L} = -\mathcal{L}_{\phi,\boldsymbol{\theta}}(\mathbf{x}) + \left\|\mathrm{sg}[\mathbf{z}_e] - \mathbf{e}_{k^*}\right\|_2^2 + \beta\left\|\mathbf{z}_e - \mathrm{sg}[\mathbf{e}_{k^*}]\right\|_2^2$$

- Use straight-through gradient estimation for encoder.

Vector Quantized VAE (VQ-VAE): Final algorithm

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- Use straight-through gradient estimation for encoder.

Sampling

- Sample $c \sim p(c) = \mathrm{Uniform}\{1, \ldots, K\}$.
- Generate $\mathbf{x} \sim p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{e}_c)$.

Outline

Likelihood-Based Models

Poor Likelihood
High-Quality Samples

$$p_1(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n} \mathcal{N}(\mathbf{x}|\mathbf{x}_i, \epsilon\mathbf{I})$$

If ϵ is very small, this model produces excellent, sharp samples but achieves poor likelihoods on test data.

Theis L., Oord A., Bethge M. A Note on the Evaluation of Generative Models, 2015

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High Likelihood
Poor Samples

$$p_2(\mathbf{x}) = 0.01p(\mathbf{x}) + 0.99p_{\text{noise}}(\mathbf{x})$$

$$\log\left[0.01p(\mathbf{x}) + 0.99p_{\text{noise}}(\mathbf{x})\right] \geq$$
$$\geq \log\left[0.01p(\mathbf{x})\right] = \log p(\mathbf{x}) - \log 100$$

This model contains mostly noisy, irrelevant samples; for high dimensions, $\log p(\mathbf{x})$ scales linearly with m.

Theis L., Oord A., Bethge M. A Note on the Evaluation of Generative Models, 2015

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$$p_1(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n}\mathcal{N}(\mathbf{x}|\mathbf{x}_i, \epsilon\mathbf{I})$$

If ϵ is very small, this model produces excellent, sharp samples but achieves poor likelihoods on test data.

High Likelihood
Poor Samples

$$p_2(\mathbf{x}) = 0.01p(\mathbf{x}) + 0.99p_{\text{noise}}(\mathbf{x})$$

$$\log\left[0.01p(\mathbf{x}) + 0.99p_{\text{noise}}(\mathbf{x})\right] \geq$$
$$\geq \log\left[0.01p(\mathbf{x})\right] = \log p(\mathbf{x}) - \log 100$$

This model contains mostly noisy, irrelevant samples; for high dimensions, $\log p(\mathbf{x})$ scales linearly with m.

- Likelihood isn't always a suitable metric for evaluating generative models.
- Sometimes, the likelihood function can't even be computed exactly.

Theis L., Oord A., Bethge M. A Note on the Evaluation of Generative Models, 2015

Likelihood-Free Learning

Motivation

We're interested in approximating the true data distribution $p_{\text{data}}(\mathbf{x})$. Instead of searching over all distributions, let's learn a model $p_{\boldsymbol{\theta}}(\mathbf{x}) \approx p_{\text{data}}(\mathbf{x})$.

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Suppose we have two sets of samples:

- $\{\mathbf{x}_i\}_{i=1}^{n_1} \sim p_{\text{data}}(\mathbf{x})$ — real data;
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Define a discriminative model (classifier):

$$p(y = 1|\mathbf{x}) = P(\mathbf{x} \sim p_{\text{data}}(\mathbf{x})); \quad p(y = 0|\mathbf{x}) = P(\mathbf{x} \sim p_{\boldsymbol{\theta}}(\mathbf{x}))$$

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Assumption

The generative model $p_{\boldsymbol{\theta}}(\mathbf{x})$ matches $p_{\text{data}}(\mathbf{x})$ if a discriminative model $p(y|\mathbf{x})$ can't distinguish between them — that is, if $p(y = 1|\mathbf{x}) = 0.5$ for every $\mathbf{x}$.

Generative Adversarial Networks (GAN)

- The more expressive the discriminator, the closer we get to the optimal $p_{\boldsymbol{\theta}}(\mathbf{x})$.
- Standard classifiers are trained by minimizing cross-entropy loss $-\mathbb{E}_{\hat{p}(\mathbf{x},y)} \log p(y|\mathbf{x})$ with $\hat{p}(\mathbf{x}, y) = \frac{1}{2}\mathbb{I}_{y=1} p_{\text{data}}(\mathbf{x}) + \frac{1}{2}\mathbb{I}_{y=0} p_{\boldsymbol{\theta}}(\mathbf{x})$.

Goodfellow I. J. et al. Generative Adversarial Networks, 2014

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Cross-Entropy for Discriminator

$$\min_{p(y|\mathbf{x})} \left[-\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log p(y=1|\mathbf{x}) - \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} \log p(y=0|\mathbf{x}) \right]$$

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Generative Model

Suppose $p_{\boldsymbol{\theta}}(\mathbf{x}, \mathbf{z}) = p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z})p(\mathbf{z})$, where $p(\mathbf{z})$ is a base distribution, and $p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) = \delta(\mathbf{x} - \mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z}))$ is deterministic.

Goodfellow I. J. et al. Generative Adversarial Networks, 2014

Generative Adversarial Networks (GAN)

Cross-Entropy for Discriminative Model

$$\max_{p(y|\mathbf{x})} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log p(y = 1|\mathbf{x}) + \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} \log p(y = 0|\mathbf{x})\right]$$

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- **Discriminator:** A classifier $p_{\phi}(y = 1|\mathbf{x}) = D_{\phi}(\mathbf{x}) \in [0, 1]$, distinguishing real and generated samples. The discriminator aims to **maximize** cross-entropy.
- **Generator:** The generative model $\mathbf{x} = \mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z})$, $\mathbf{z} \sim p(\mathbf{z})$, seeks to fool the discriminator. The generator aims to **minimize** cross-entropy.

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GAN Objective

$$\min_G \max_D \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} \log(1 - D(\mathbf{x}))\right]$$

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Generative Adversarial Networks (GAN)

Cross-Entropy for Discriminative Model

$$\max_{p(y|\mathbf{x})} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log p(y=1|\mathbf{x}) + \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} \log p(y=0|\mathbf{x})\right]$$

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Goodfellow I. J. et al. Generative Adversarial Networks, 2014

Summary

- ELBO surgery gives insights into the prior's influence in VAEs; the optimal prior is the aggregated variational posterior.
- The mismatch between $p(\mathbf{z})$ and $q_{\text{agg},\phi}(\mathbf{z})$ is widely regarded as the principal reason for VAE-generated image blurriness.
- Vector quantization provides a way to construct VAEs with discrete latents and deterministic variational posteriors.
- The straight-through gradient estimator allows gradients to pass as if quantization were an identity operation during backpropagation.
- Likelihood is not always a suitable metric for evaluating generative models
- Likelihood-free learning motivates the use of discriminative models to match distributions.